

FILE : CH8SOL1

```
if (4+5==10):  
    print("true")  
else:  
    print("False")  
print("True")
```

FILE : CH8SOL2

```
name=""  
while True :    # while True:(boolean literal) creates an  
infinite loop  
    name=input("enter name('end' to exit):")  
    if name == "end":    # to break the infinite looping  
        break  
    print("Hello",name)  
else:  
    print("wasn't it fun ? ")
```

#2

```
print()  
print("#2")  
name=""  
while name != "end":  
    name=input("enter name('end' to exit):")  
    if name == "end":  
        pass  
    print("Hello",name)  
else:  
    print("wasn't it fun ? ")
```

#3

```
print()  
print("#3")  
name=""  
while 5 : #while 5: creates infinite loop for any non zero value  
    name=input("enter name('end' to exit):")  
    if name == "end":    # to break the infinite looping  
        break  
    print("Hello",name)  
else:  
    print("wasn't it fun ? ")
```

FILE : CH8SOL3

```
N =int(input('enter number: '))
sum=0
i=1
while i<=N :
    sum=sum+i
    i=i+2
print(sum)
```

FILE : CH8SOL4

```
N=int(input("Enter N: "))
i=1
sum=0
while i< N:
    if i%2==0:
        sum=sum+i
    i=i+1
print(sum)
```

FILE : CH8SOL4A

```
number= int(input("enter number:"))
n,result = number,1
while True or n:
    result = result *n
    n=n-1
print("factorial of",number,"is",result)
```

FILE : CH8SOL5

to find the largest among 3 numbers

```
num1=float(input("Enter 1st number: "))
num2=float(input("Enter 2nd number: "))
num3=float(input("Enter 3rd number: "))
```

```

if (num1 > num2) and(num1 > num3):
    largest=num1
elif (num2 > num1) and(num2 > num3):
    largest=num2
else:
    largest=num3
print("largest number is: ",largest)
if (num1 < num2) and(num1 < num3):
    smallest=num1
elif (num2 < num1) and(num2 < num3):
    smallest=num2
else:
    smallest=num3
print("smallest among 3 numbers is : ",smallest)
FILE : CH8SOL6

```

```

# to break up a 6 digit into 3 two digit numbers

```

```

num=int(input("Enter 6 digit number: "))

```

```

if num<100000 or num>999999:

```

```

    print("please enter 6 digit: ")

```

```

else:

```

```

    n1=num%100

```

```

    # last 2 digits of the number

```

```

    int1=num//100

```

```

    n2=int1%100

```

```

    # middle 2 digits of the number

```

```

    int2=int1//100

```

```

    n3=int2%100

```

```

    # first 2 digits of the number

```

```

    print("Three 2-digit number are",n3,n2,n1)

```

```

print();print("#2")

```

```

# to find the first, middle and last digit of a number

```

```

import math

```

```

num=int(input("Enter a number: "))

```

```

if len(str(num))%2 == 0:

```

```

    print("please enter odd number of digits : ")

```

```

else:

```

```

    n1=num%10

```

```

    # last digit of the number

```

```

    n2=str(num)[len(str(num))//2]

```

```

    #middle digit of the number

```

```

    n3=num//pow(10,len(str(num))-1)

```

```

    # first digit of the number

```

```

    print("first digit is :",n3)

```

```

    print("middle digit is : " ,n2)

```

```

    print("last digit is :",n1)

```

FILE : CH8SOL7

```
import math
num=int(input("Enter number: "))
ln= len(str(num))
lst=num%10
fst=num// math.pow(10,ln-1)
print("length of given number",num,"is",ln)
print("The first digit is",fst)
print("The last digit is",lst)
print("first digit rising to the length:",\
math.pow(fst,ln))
print("last digit rising to the length:",\
math.pow(lst,ln))
```

FILE : CH8SOL8

```
small=smallest=0
for i in range(5):
    n=int(input("Enter number: "))
    if i == 0:
        small = n
    elif i == 1:
        if n<= small:
            smaller =n
        else:
            smaller= small
            small=n
    else:
        if n < smaller:
            small= smaller
            smaller=n
        elif n< small:
            small=n
print("The lowest number is: ",smaller)
print("The second lowest number is: ",small)
```

FILE : CH8SOL8A

to print lowest and 2nd lowest no.

```
small=smallest=0
for i in range(5):
    n=int(input("Enter number: "))
    if i == 0:
        small = n
    elif n< small:
        small=n
for i in range(5):
    n=int(input("Enter number: "))
    if n< small:
        pass
    else:
        small=n""
print("The lowest number is: ",small)
#print("The second lowest number is: ",small)
```

FILE : CH8SOL8B

to print lowest, 2nd lowest, largest and 2nd largest nos.

```
small=smallest=0
for i in range(5):
    n=int(input("Enter number: "))
    if i == 0:
        small = n
        s2=n
    elif n< small:
        s2=small
        small=n
    if i == 0:
        lar = n
        l2=n
    elif n> lar:
        l2=lar
        lar=n
```

```
print("The lowest number is: ",small)
print("The second lowest number is: ",s2)
print("The largest number is: ",lar)
print("The second largest number is: ",l2)
```

FILE : CH8SOL9

```
import random
secretNum= random.randint(1,100)
guessNumString=int(input("Guess a number between 1 and 100
inclusive: "))
while guessNumString !=secretNum:
    if guessNumString<secretNum:
        print("your guess is too low")
    else:
        print("your guess is too high")
    guessNum=int(input("Guess a number between 1 and 100
inclusive: "))
print("congratulation ! You guess the correct number")
```

FILE : CH8SOL10

```
first=0; second=1
print(first); print(second)
for a in range(1,19):
    third=first+second
    print(a+2,"  ",third)
    first,second=second,third
```

FILE : CH8SOL11

```
num=int(input("enter an integer(>1000)  the last digit must not
be 0 :"))
tnum= num; reverse=0
while tnum:
    digit = tnum%10
    tnum=int(tnum/10)
    reverse= reverse*10+digit
print("Reverse  of",num,"is",reverse)
```

FILE : CH8SOL11A

```
num=int(input("enter an integer(>1000): "))
tnum= num
reverse=0
for i in range(tnum,0,-1) :
    digit = tnum%10
    print(digit)
    tnum=int(tnum/10)
    reverse= reverse*10+digit
print("Reverse  of",num,"is",reverse)
```

FILE : CH8SOL12

```
num=int(input("Enter an intger: "))
mid=num//2
print("the factor of ", num," other than 1 and itself are:" )
for a in range(2,mid+1):
    if num%a==0:
        print(a,end=' ')
else:
    print("-End-")
```

FILE : CH8SOL13

```
angle1=angle2=angle3=0
angle1=float(input("enter angle1: "))
angle2=float(input("enter angle2: "))
angle3=float(input("enter angle3: "))
if (angle1+angle2+angle3)==180:
    print("angle form a triangle")
else:
    print("angle do not form a triangle")
```

FILE : CH8SOL14

```
x=int(input("enter a positive number: "))
n=int(input("enter a power(n): "))
power=1
for r in range(n):
    power=power*x
print(x,"to the power",n,"is",power)
```

FILE : CH8SOL15

```
num=int(input("enter a number:"))
fac=1
for i in range(num,0,-2):
    fac*=i
print(num,"!! is :",fac)
```

FILE : CH8SOL16

```
print("first 10 mersenne number are: ")
for a in range(1,11):
    mersnum=2**a-1
    print(mersnum, end=" ")
print()
```

FILE : CH8SOL17

```
for a in range(3,10,2):
    for b in range(1,a,2):
        print(b,end='')
    print()
```

FILE : CH8SOL18

```
for a in range(1,21):
    mersnum=2**a-1
    mid=int(mersnum/2)
    for b in range(2,mid):
        if mersnum % b ==0:
            print(mersnum)
            break
    else:
        print(mersnum, "\tprime")
```

FILE : CH8SOL19

```
weight_in_kg=float(input("enter weight in Kg: "))
height_in_meter=float(input("enter height in meter: "))
bmi=weight_in_kg/(height_in_meter*height_in_meter)
print("BMI is :",bmi,end=" ")
if bmi <18.5:
    print("...under weight")
elif bmi<25:
    print("...normal")
elif bmi<30:
    print("...over weight")
else:
    print("...obese")
```

FILE : CH8SOL20

```
x=float(input("enter value of x: "))
n=int(input("enter value of n (for x**n): "))
s=0
for a in range (n+1):
    s+=x**a
print("sum of first",n,"terms :",s)
```

FILE : CH8SOL21

```
x=int(input("enter value of x: "))
n=int(input("enter power of(n): "))
s=0
sign=+1
for a in range(n+1):
    term=(x**a)*sign
    s+=term
    sign*=-1
print("sum of first", n, "terms: ",s)
```

FILE : CH8SOL22

```
x=int(input("enter value of x: "))
n=int(input("enter power of(n): "))
s=x
sign=+1
for a in range(2,n+1):
    f=1
    for i in range(2,a+1):
        f*=i
    term=((x**a)*sign)/f
    s+=term
    sign*=-1
print("sum of first",n,"terms : ",s)
```

FILE : CH8SOL23

```
n=int(input("Enter number: "))
summ=0
for i in range(1,n):
    if (n%i==0):
        summ=summ+i
if (summ==n):
    print("the number ",n,"is a perfect number!")
else:
    print("the number",n,"is not a perfect number!")
```

FILE : CH8SOL24

```
num= int(input("enter 3 digit number:"))
summ=0
temp=num
while (temp>0):
    digit =temp%10
    summ+=digit**3
    temp//=10
if (num==summ):
    print(num,"is an armstrong number :")
else:
    print(num,"is not an armstrong number :")
```

FILE : CH8SOL25

```
num=int(input("enter number:"))
wnum=num
rev=0
while(wnum>0):
    dig=wnum%10
    rev=rev*10+dig
    wnum=wnum//10
if (num==rev):
    print("number",num,"is a palindrome!")
else:
    print("number",num,"is not a palindrome!")
```

FILE : CH8SOL26

```
t=int(input("enter many terms? (enter 2+ value) :"))
first=0; second=1
print("\nFibonacci series is : ")
print(first,end=",")
print(second,end=",")
for i in range(2,t):
    next =first+ second
    print( end=",",next)
    first= second
    second=next
```

FILE : CH8SOL27A

```
x=int(input("enter the number :"))
y=int(input("enter the number :"))
if x>y:
    smaller=y
else:
    smaller=x
for i in range(1,smaller+1):
    if ((x%i==0)and(y%i==0)):
        hcf=i
lcm=(x*y)/hcf
print("the H.C.F of",x,"and",y,"is",hcf)
print("the L.C.M of",x,"and",y,"is",lcm)
```

FILE : CH8SOL28

```
sum=0
n=int(input("how many terms?"))
#added 2 to n becasuse started with 2
for a in range(2,n+2):
    t=0
    for b in range (1,a):
        t+=b
    print("Term",(a-1),":",t)
    sum+=t
print("sum of",n,"term is", sum)
```

FILE : CH8SOL29

```
import random
n=int(input("how many dice throws ?"))
for i in range(1,n+1):
    print("throw",i,":",random.randint(1,6))
```

FILE : CH8SOL30

```
n=1
for a in range(5):
    print(n)
    n=n*10+1          #number generator
```

FILE : CH8SOL31

```
for i in range(4):
    for j in range(4,i,-1):
        print(j,end=" ")
    else:
        print()
```